Saviours AI

ICSE CLASS 10 · 2027 SYLLABUS

Chemistry Cheat Sheet

Every reaction, equation and test from ICSE Class 10 Chemistry. 8 pages. Nothing missing.

Periodic Table | Bonding | Acids & Salts | Analytical | Mole Concept | Electrolysis | Metallurgy | HCl · NH_3 · HNO_3 · H_2SO_4 | Organic

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01 Periodic table & trends

Periodic properties: the trends

Property	Down a group	Across a period →
Atomic size	Increases (more shells)	Decreases (more nuclear charge)
Metallic character	Increases	Decreases
Non-metallic character	Decreases	Increases
Ionisation potential	Decreases	Increases
Electron affinity	Decreases	Increases
Electronegativity	Decreases	Increases

- **Ionisation potential:** energy needed to remove an electron from an isolated gaseous atom.
- **Electron affinity:** energy released when an atom gains an electron.
- **Electronegativity:** tendency of an atom to attract the shared pair of electrons.

Alkali metals vs halogens

	Alkali metals (Gp 1)	Halogens (Gp 17)
Valence e^-	1	7
Ion formed	M^+ (loses e^-)	X^- (gains e^-)
Reactivity down group	Increases	Decreases
Nature of oxide / acid	Basic	Acidic
State	Soft metals	F_2, Cl_2 gas; Br_2 liquid; I_2 solid
Colour	Silvery	F pale yellow; Cl green; Br red-brown; I violet

- **Why K is more reactive than Na:** larger atom, outer electron is further from the nucleus and is lost more easily.
- **Why F is more reactive than Cl:** smaller atom, higher electronegativity, gains an electron more easily.

02 Chemical bonding

Electrovalent (ionic) bond

- Formed by **transfer** of electrons: metal loses, non-metal gains. Each atom reaches a stable duplet or octet.

$$Na\ (2,8,1) \rightarrow Na^+\ (2,8) + e^-$$

$$Cl\ (2,8,7) + e^- \rightarrow Cl^-\ (2,8,8)$$

NaCl — $[Na]^+\ [:\ddot{Cl}:]^-$

$MgCl_2$ — $[Mg]^{2+}\ 2\ [:\ddot{Cl}:]^-$

CaO — $[Ca]^{2+}\ [:\ddot{O}:]^{2-}$

Covalent bond

- Formed by **sharing** of electrons between non-metals.

Molecule	Bond	Electron dot idea
H_2	Single	H:H
Cl_2	Single	Cl–Cl, 3 lone pairs each
O_2	Double	O=O, 2 lone pairs each
N_2	Triple	N≡N, 1 lone pair each
CH_4	4 single	C with 4 shared pairs
CCl_4	4 single	C with 4 Cl
CO_2	2 double	O=C=O
HCl	Single, polar	H–Cl
NH_3	3 single, 1 lp	N with 3 H, 1 lone pair
H_2O	2 single, 2 lp	O with 2 H, 2 lone pairs

- **Non-polar:** equal sharing (H_2, Cl_2, O_2, N_2, CH_4, CCl_4). **Polar:** unequal sharing, due to electronegativity difference (HCl, NH_3, H_2O).

Coordinate (dative) bond

- Both electrons of the shared pair come from **one atom** (the donor). It has a **lone pair**.

Hydronium ion — $H_2O + H^+ \rightarrow H_3O^+$

Ammonium ion — $NH_3 + H^+ \rightarrow NH_4^+$

- **Donor:** O in water, N in ammonia. **Acceptor:** H^+ (no electrons).
- **Arrow notation:** donor → acceptor, e.g., $H_3N \rightarrow H^+$.

Electrovalent vs covalent compounds

Property	Electrovalent	Covalent
State	Hard, crystalline solids	Gases, liquids, soft solids
M.P. and B.P.	High	Low
Solubility	Soluble in water, insoluble in organic solvents	Insoluble in water (mostly), soluble in organic solvents
Conductivity	Conduct when molten or in solution	Do not conduct
Reaction speed	Fast (ionic)	Slow (molecular)
Bond	Strong electrostatic	Weak intermolecular forces

03 Acids, bases & salts

Indicators & pH

Indicator	Acid	Neutral	Alkali
Litmus	Red	Purple	Blue
Methyl orange	Red / pink	Orange	Yellow
Phenolphthalein	Colourless	Colourless	Pink
Universal	Red–orange	Green	Blue–violet

pH < 7 acid · pH = 7 neutral · pH > 7 alkali

pH of pure water — 7

pH of lemon / vinegar — 2–3

pH of HCl (strong) — 1

pH of NaOH (strong) — 13–14

Definitions & ionisation

- **Acid:** gives H^+ (as H_3O^+) as the only positive ion in water.
- **Alkali:** gives OH^- as the only negative ion in water. **Base:** oxide or hydroxide of a metal that reacts with an acid to give salt and water.

Hydrochloric	$HCl \rightarrow H^+ + Cl^-$
Sulphuric	$H_2SO_4 \rightarrow 2H^+ + SO_4^{2-}$
Nitric	$HNO_3 \rightarrow H^+ + NO_3^-$
Ethanoic (weak)	$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$
Sodium hydroxide	$NaOH \rightarrow Na^+ + OH^-$
Ammonium hydroxide	$NH_4OH \rightleftharpoons NH_4^+ + OH^-$

- **Basicity** of an acid = number of H^+ per molecule: HCl (1), H_2SO_4 (2), H_3PO_4 (3).
- **Acidity** of a base = number of OH^-: NaOH (1), $Ca(OH)_2$ (2).

Types of salts

Type	Meaning	Example
Normal	All H^+ replaced	NaCl, Na_2SO_4
Acid	Some H^+ still replaceable	$NaHSO_4$, $NaHCO_3$
Basic	Contains OH^- or O^{2-}	Basic lead chloride Pb(OH)Cl
Double	Two cations or anions	Potash alum
Complex	Contains complex ion	$Na[Al(OH)_4]$, $[Cu(NH_3)_4]SO_4$

- **Salt:** formed by partial or complete replacement of the H^+ of an acid by a metal or by NH_4^+.

Action of dilute acids on salts

Carbonate	$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2\uparrow$
Bicarbonate	$NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2\uparrow$
Sulphite	$Na_2SO_3 + 2HCl \rightarrow 2NaCl + H_2O + SO_2\uparrow$
Bisulphite	$NaHSO_3 + HCl \rightarrow NaCl + H_2O + SO_2\uparrow$
Sulphide	$FeS + H_2SO_4 \rightarrow FeSO_4 + H_2S\uparrow$

- CO_2 turns limewater milky. SO_2 smells of burning sulphur and turns acidified dichromate green. H_2S smells of rotten eggs, turns lead acetate paper black.

Six methods to prepare salts

Method	Example
Direct combination	$Fe + S \rightarrow FeS$; $2Fe + 3Cl_2 \rightarrow 2FeCl_3$
Displacement	$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2\uparrow$
Precipitation (double decomp.)	$AgNO_3 + NaCl \rightarrow AgCl\downarrow + NaNO_3$
Neutralisation of insoluble base	$CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$
Neutralisation of an alkali (titration)	$NaOH + HCl \rightarrow NaCl + H_2O$
Acid + carbonate	$CuCO_3 + H_2SO_4 \rightarrow CuSO_4 + H_2O + CO_2\uparrow$

04 Analytical chemistry

Colour of salts & solutions

Salt	Solid	Solution
Copper(II) sulphate	Blue (hydrated)	Blue
Iron(II) sulphate	Pale green	Pale green
Iron(III) chloride	Brown / yellow	Yellow-brown
Zinc sulphate	White	Colourless
Lead nitrate	White	Colourless
Ammonium chloride	White	Colourless

Add NaOH drop by drop, then in excess

Cation	Few drops	In excess
Fe^{2+}	Dirty green ppt. $Fe(OH)_2$	Insoluble
Fe^{3+}	Reddish-brown ppt. $Fe(OH)_3$	Insoluble
Cu^{2+}	Pale blue ppt. $Cu(OH)_2$	Insoluble
Zn^{2+}	White ppt. $Zn(OH)_2$	Dissolves → sodium zincate
Pb^{2+}	White ppt. $Pb(OH)_2$	Dissolves → sodium plumbite

Fe^{2+}	$FeSO_4 + 2NaOH \rightarrow Fe(OH)_2\downarrow + Na_2SO_4$
Cu^{2+}	$CuSO_4 + 2NaOH \rightarrow Cu(OH)_2\downarrow + Na_2SO_4$
Zn^{2+}	$ZnSO_4 + 2NaOH \rightarrow Zn(OH)_2\downarrow + Na_2SO_4$
$Zn(OH)_2$ excess	$Zn(OH)_2 + 2NaOH \rightarrow Na_2ZnO_2 + 2H_2O$

Add NH_4OH drop by drop, then in excess

Cation	Few drops	In excess
Fe^{2+}	Dirty green ppt.	Insoluble
Fe^{3+}	Reddish-brown ppt.	Insoluble
Cu^{2+}	Pale blue ppt.	**Dissolves → inky blue** solution
Zn^{2+}	White ppt.	Dissolves → colourless
Pb^{2+}	White ppt.	Insoluble

Inky blue	$Cu(OH)_2 + 4NH_3 \rightarrow [Cu(NH_3)_4]^{2+} + 2OH^-$

Special tests

Ammonium salt + NaOH	$NH_4Cl + NaOH \rightarrow NaCl + H_2O + NH_3\uparrow$

- NH_3 has a pungent smell and turns moist red litmus **blue**. It gives dense white fumes with HCl.
- Cl^- **test:** + $AgNO_3$ → white curdy ppt. of AgCl, soluble in NH_4OH.
- SO_4^{2-} **test:** + $BaCl_2$ → white ppt. of $BaSO_4$, insoluble in acids.
- CO_3^{2-} **test:** + dilute acid → CO_2 → limewater turns milky.

Gas laws

- **Gay-Lussac:** gases react in simple whole-number volume ratios, at the same T and P.

$N_2 + 3H_2 \rightarrow 2NH_3$: 1 vol : 3 vol → 2 vol

- **Avogadro:** equal volumes of all gases at the same T and P contain **equal numbers of molecules**.
- **Atomicity:** H_2, O_2, N_2, Cl_2 are diatomic (2); O_3 is triatomic; P_4 is 4; S_8 is 8; noble gases are monoatomic.

Amphoteric substances

- React with **both acids and alkalis** to give salt and water.
- **Amphoteric metals:** Al, Zn, Pb. **Oxides:** Al_2O_3, ZnO, PbO. **Hydroxides:** $Al(OH)_3$, $Zn(OH)_2$, $Pb(OH)_2$.

Zinc + alkali	$Zn + 2NaOH \rightarrow Na_2ZnO_2 + H_2\uparrow$
Aluminium + alkali	$2Al + 2NaOH + 2H_2O \rightarrow 2NaAlO_2 + 3H_2\uparrow$
Al_2O_3 + alkali	$Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$
ZnO + alkali	$ZnO + 2NaOH \rightarrow Na_2ZnO_2 + H_2O$
PbO + alkali	$PbO + 2NaOH \rightarrow Na_2PbO_2 + H_2O$

05 Mole concept & stoichiometry

Mole: three-way relation

$$n = m / M = V / 22.4 = N / (6.022 \times 10^{23})$$

n	number of moles
m / M	mass (g) / molar mass (g mol^{-1})
V	volume of gas at STP (litres)
Molar volume	22.4 L at STP
Avogadro number	6.022×10^{23} per mole
STP	0 °C (273 K) and 1 atm

- **1 mole** of any substance contains 6.022×10^{23} particles.
- **Gram atomic mass** = mass of 1 mole of atoms. **Gram molecular mass** = mass of 1 mole of molecules.

Vapour density

$$V.D. = \text{molar mass of gas} / \text{molar mass of } H_2$$

$$\text{Molecular mass} = 2 \times V.D.$$

Example: V.D. of CO_2 = 44/2 = **22**.

Density of gas	mass / volume
Relative molecular mass	sum of atomic masses

Empirical & molecular formula

- **Empirical formula:** simplest whole-number ratio of atoms. **Molecular formula:** actual number of atoms.

Step	Action
1	Write % (or masses) of each element
2	Divide by atomic mass → moles
3	Divide by the smallest → ratio
4	Round / multiply to whole numbers → **empirical formula**
5	n = molecular mass / empirical formula mass
6	Molecular formula = $(\text{empirical})_n$

Percentage: (mass of element / molar mass) × 100

Stoichiometry: calculations from equations

- **1.** Balance the equation. **2.** Convert the given to moles. **3.** Use the mole ratio. **4.** Convert to the required unit.

Example: $2H_2 + O_2 \rightarrow 2H_2O$. 4 g H_2 = 2 mol → needs 1 mol O_2 (22.4 L at STP) → gives 2 mol H_2O (36 g).

Volumes of gases at the same T and P are in the same ratio as moles.

06 Electrolysis

Terms & electrolytes

- **Electrolysis:** chemical decomposition of an electrolyte by passing electric current.
- **Anode** (+): oxidation, loses electrons. **Cathode** (−): reduction, gains electrons. **Anions** go to the anode, **cations** to the cathode.

Type	Examples
Strong electrolytes	HCl, H_2SO_4, HNO_3, NaOH, KOH, NaCl, $CuSO_4$
Weak electrolytes	H_2CO_3, CH_3COOH, NH_4OH
Non-electrolytes	Sugar, alcohol, glucose, urea, distilled water

- **Contain ions only:** NaCl, KOH. **Molecules only:** sugar. **Both:** weak acids, e.g., CH_3COOH.

Selective discharge

- **Cations:** the one **lower in the reactivity series** discharges first. Order: $Ag^+ > Cu^{2+} > H^+ > Pb^{2+} > Fe^{2+} > Zn^{2+} > Al^{3+} > Mg^{2+} > Ca^{2+} > Na^+ > K^+$.
- **Anions:** OH^- discharges before SO_4^{2-} and NO_3^-. Cl^- is discharged in concentrated solutions.
- **Electrode nature:** active (Cu, Ag, Ni) electrodes may take part. Inert (Pt, C) do not.
- **Concentration** also decides: high concentration favours discharge.

The four standard electrolyses

Electrolyte / electrodes	Cathode (−)	Anode (+)
Molten $PbBr_2$, inert	$Pb^{2+} + 2e^- \rightarrow Pb$ (grey bead)	$2Br^- \rightarrow Br_2 + 2e^-$ (red-brown fumes)
Acidified water, Pt	$2H^+ + 2e^- \rightarrow H_2\uparrow$	$4OH^- \rightarrow 2H_2O + O_2 + 4e^-$ ($H_2 : O_2 = 2 : 1$)
$CuSO_4$(aq), Pt / C	$Cu^{2+} + 2e^- \rightarrow Cu$ (red-brown deposit)	$4OH^- \rightarrow 2H_2O + O_2 + 4e^-$ (blue colour fades)
$CuSO_4$(aq), Cu	$Cu^{2+} + 2e^- \rightarrow Cu$	$Cu \rightarrow Cu^{2+} + 2e^-$ (anode dissolves; blue stays)

Electro-refining of copper

- **Anode:** impure copper. **Cathode:** pure copper strip. **Electrolyte:** acidified $CuSO_4$.

Anode	$Cu \rightarrow Cu^{2+} + 2e^-$
Cathode	$Cu^{2+} + 2e^- \rightarrow Cu$

- Impurities fall as **anode mud**. Copper moves from anode to cathode.

Electroplating

- **Cathode:** object to be plated. **Anode:** pure metal of the coating. **Electrolyte:** salt solution of the plating metal.

Plating	Electrolyte	Anode	Cathode reaction
Silver	Sodium argentocyanide $Na[Ag(CN)_2]$	Pure silver	$Ag^+ + e^- \rightarrow Ag$
Nickel	Nickel sulphate solution	Pure nickel	$Ni^{2+} + 2e^- \rightarrow Ni$

- **Why plate?** To prevent corrosion, improve appearance, increase hardness.
- **Conditions:** low current, steady, low-concentration electrolyte gives smooth deposition.

07 Metallurgy

Reactivity series

K > Na > Ca > Mg > Al > Zn > Fe > Pb > (H) > Cu > Hg > Ag > Au

- **Top (K, Na, Ca, Mg, Al):** extracted by **electrolysis** of molten compounds.
- **Middle (Zn, Fe, Pb):** reduced by **carbon / CO / H_2**.
- **Low (Cu, Ag, Au):** found native or reduced easily.

Ores of key metals

Metal	Ore	Formula
Aluminium	Bauxite	$Al_2O_3 \cdot 2H_2O$
Aluminium	Cryolite	Na_3AlF_6
Iron	Haematite	Fe_2O_3
Iron	Magnetite	Fe_3O_4
Iron	Iron pyrites	FeS_2
Zinc	Zinc blende	ZnS
Zinc	Calamine	$ZnCO_3$

- **Mineral:** naturally occurring compound of a metal. **Ore:** a mineral from which the metal is extracted **profitably**.

Stages of extraction

- **1. Dressing:** crushing, then **hydraulic washing** (heavy ore), **magnetic separation** (magnetic ore), **froth flotation** (sulphide ores).
- **2. Conversion to oxide:**

Calcination (carbonates) $ZnCO_3 \rightarrow ZnO + CO_2$

Roasting (sulphides) $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$

- **3. Reduction:**

Zinc $ZnO + C \rightarrow Zn + CO$

Iron $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

Copper $CuO + H_2 \rightarrow Cu + H_2O$

- **Not reduced by carbon:** Al_2O_3, MgO. They need electrolysis.

Aluminium: Baeyer's process (purifying bauxite)

Digestion $Al_2O_3 \cdot 2H_2O + 2NaOH \rightarrow 2NaAlO_2 + 3H_2O$

Dilution $NaAlO_2 + 2H_2O \rightarrow Al(OH)_3\downarrow + NaOH$

Heating $2Al(OH)_3 \rightarrow Al_2O_3 + 3H_2O$

- **Impurities:** Fe_2O_3 (does not dissolve in NaOH) is filtered off as red mud.

Hall–Héroult: electrolytic extraction

- **Electrolyte:** pure Al_2O_3 dissolved in molten **cryolite** (Na_3AlF_6) with **fluorspar** (CaF_2), at ~950 °C.
- **Purpose:** cryolite lowers the melting point (2050 °C → ~950 °C) and increases conductivity. Fluorspar lowers it further.
- **Cathode:** graphite-lined steel tank. **Anode:** graphite rods.

Cathode $Al^{3+} + 3e^- \rightarrow Al$ (sinks, tapped off)

Anode $2O^{2-} \rightarrow O_2 + 4e^-$

Anode wears away $C + O_2 \rightarrow CO_2$ (replaced regularly)

Alloys & amalgam

Alloy	Composition	Use
Stainless steel	Fe, Cr, Ni, C	Utensils, surgical tools
Brass	Cu 80, Zn 20	Utensils, taps
Bronze	Cu 90, Sn 10	Statues, medals
Duralumin	Al, Cu, Mg, Mn	Aircraft
Magnalium	Al 95, Mg 5	Balance beams
Fuse metal / solder	Pb, Sn	Fuses, soldering
Amalgam	Hg with a metal	Dental fillings

- **Alloy:** homogeneous mixture of a metal with other metals or non-metals. Harder, resistant to corrosion.

08 Study of compounds: HCl & NH_3

Hydrogen chloride: preparation

Lab method $NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl\uparrow$ (below 200 °C)

Higher heat $NaHSO_4 + NaCl \rightarrow Na_2SO_4 + HCl$ (above 200 °C)

- **Dry with** conc. H_2SO_4. **Collect by** upward delivery (**heavier** than air). Not collected over water.
- **Fountain experiment:** very soluble in water; red fountain in blue litmus.
- **Preparing HCl solution:** use an inverted funnel just touching water, which avoids **back suction**.

Ammonia: preparation

Lab $2NH_4Cl + Ca(OH)_2 \rightarrow CaCl_2 + 2H_2O + 2NH_3\uparrow$

From nitrides $Mg_3N_2 + 6H_2O \rightarrow 3Mg(OH)_2 + 2NH_3$

Aluminium nitride $AlN + 3H_2O \rightarrow Al(OH)_3 + NH_3$

Ammonium salt + alkali $(NH_4)_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O + 2NH_3$

- **Dry with** quicklime CaO. **Collect by** downward displacement of air (lighter than air).

HCl: properties & tests

With metals	$Zn + 2HCl \rightarrow ZnCl_2 + H_2\uparrow$
With oxides	$CuO + 2HCl \rightarrow CuCl_2 + H_2O$
With carbonates	$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$
With ammonia	$NH_3 + HCl \rightarrow NH_4Cl$ (dense white fumes)
With $AgNO_3$	$HCl + AgNO_3 \rightarrow AgCl\downarrow + HNO_3$
With $Pb(NO_3)_2$	$Pb(NO_3)_2 + 2HCl \rightarrow PbCl_2\downarrow + 2HNO_3$
Aqua regia	3 conc. HCl : 1 conc. HNO_3

- **Aqua regia** dissolves gold and platinum.

Haber's process

$$N_2 + 3H_2 \rightleftharpoons 2NH_3 + \text{heat}$$

Catalyst	finely divided iron
Promoter	Mo or Al_2O_3
Temperature	450–500 °C
Pressure	200–900 atm

- **Le Chatelier:** high P and low T favour NH_3, but a moderate T gives a faster rate.

Ammonia: reactions

Burning in O_2	$4NH_3 + 3O_2 \rightarrow 2N_2 + 6H_2O$ (yellow flame)
Catalytic oxidation	$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$ (Pt, 800 °C)
With CuO	$3CuO + 2NH_3 \rightarrow 3Cu + N_2 + 3H_2O$
With PbO_2	$3PbO_2 + 4NH_3 \rightarrow 3Pb + 2N_2 + 6H_2O$
Excess NH_3 + Cl_2	$8NH_3 + 3Cl_2 \rightarrow 6NH_4Cl + N_2$
Excess Cl_2 + NH_3	$NH_3 + 3Cl_2 \rightarrow NCl_3 + 3HCl$
With acids	$2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$

- **Solution (NH_4OH):** $FeCl_3$ → $Fe(OH)_3$ (brown), $FeSO_4$ → $Fe(OH)_2$ (green), $Pb(NO_3)_2$ → $Pb(OH)_2$ (white), $CuSO_4$ → pale blue then inky blue.
- **Uses:** fertilisers, nitric acid, explosives, refrigerant, cleaning agents.

09 Study of compounds: HNO_3 & H_2SO_4

Nitric acid: preparation

Lab	$KNO_3 + H_2SO_4 \rightarrow KHSO_4 + HNO_3$ (below 200 °C)

- **Apparatus:** all-glass retort with water-cooled receiver. The acid is yellow because of dissolved NO_2.
- **Ostwald's process:**

Step 1	$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$ (Pt gauze, 800 °C)
Step 2	$2NO + O_2 \rightarrow 2NO_2$
Step 3	$4NO_2 + 2H_2O + O_2 \rightarrow 4HNO_3$

Nitric acid: properties

Decomposes	$4HNO_3 \rightarrow 4NO_2 + 2H_2O + O_2$
Cu + conc.	$Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$
Cu + dilute	$3Cu + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$
Carbon	$C + 4HNO_3 \rightarrow CO_2 + 4NO_2 + 2H_2O$
Sulphur	$S + 6HNO_3 \rightarrow H_2SO_4 + 6NO_2 + 2H_2O$
Acid: Mg + dilute	$Mg + 2HNO_3 \rightarrow Mg(NO_3)_2 + H_2$

- **Brown ring test** for nitrate: $FeSO_4$ + conc. H_2SO_4 down the side of the tube → brown ring.

Sulphuric acid: Contact process

Step 1	$S + O_2 \rightarrow SO_2$ (or $4FeS_2 + 11O_2 \rightarrow 2Fe_2O_3 + 8SO_2$)
Step 2	$2SO_2 + O_2 \rightleftharpoons 2SO_3$ + heat
Conditions	V_2O_5 catalyst, 450 °C, 1–2 atm
Step 3	$SO_3 + H_2SO_4 \rightarrow H_2S_2O_7$ (oleum)
Step 4	$H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$

- **Why not dissolve SO_3 in water directly?** It makes a fine mist (acid fog), hard to condense.

Dilute H_2SO_4: acid reactions

Metal	$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
Oxide	$CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$
Carbonate	$Na_2CO_3 + H_2SO_4 \rightarrow Na_2SO_4 + H_2O + CO_2$
Sulphite	$Na_2SO_3 + H_2SO_4 \rightarrow Na_2SO_4 + H_2O + SO_2$
Sulphide	$ZnS + H_2SO_4 \rightarrow ZnSO_4 + H_2S$

- **Dibasic acid:** gives two series of salts: $NaHSO_4$ (acid salt) and Na_2SO_4 (normal salt).

Conc. H_2SO_4: other properties

- **Oxidising agent:**

Carbon	$C + 2H_2SO_4 \rightarrow CO_2 + 2SO_2 + 2H_2O$
Sulphur	$S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$

- **Dehydrating agent:**

Sugar	$C_{12}H_{22}O_{11} \rightarrow 12C + 11H_2O$ (black charred mass)
$CuSO_4 \cdot 5H_2O$	blue → white (anhydrous)

- **Non-volatile:** displaces volatile acids from salts.

With NaCl	$NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$
With KNO_3	$KNO_3 + H_2SO_4 \rightarrow KHSO_4 + HNO_3$

10 Organic chemistry

Carbon basics

- **Tetravalency:** carbon forms four bonds. **Catenation:** carbon bonds with itself to form chains and rings.
- **Chains:** straight, branched, cyclic. **Benzene:** C_6H_6, a ring of 6 carbons with alternate double bonds.
- **Saturated:** only single bonds (alkanes). **Unsaturated:** double or triple bond (alkenes, alkynes).

Homologous series

Series	General formula	Suffix
Alkane	C_nH_{2n+2}	-ane
Alkene	C_nH_{2n}	-ene
Alkyne	C_nH_{2n-2}	-yne

- **Same functional group**, same general formula, successive members differ by **CH_2 (14 u)**, gradual change in physical properties, similar chemical properties.

C	Alkane	Alkene	Alkyne
1	Methane CH_4	—	—
2	Ethane C_2H_6	Ethene C_2H_4	Ethyne C_2H_2
3	Propane C_3H_8	Propene C_3H_6	Propyne C_3H_4
4	Butane C_4H_{10}	Butene C_4H_8	Butyne C_4H_6
5	Pentane C_5H_{12}	Pentene C_5H_{10}	Pentyne C_5H_8

Functional groups & naming

Group	Formula	Suffix / prefix	Example
Halide	–X	chloro-, bromo-	Chloromethane
Alcohol	–OH	-ol	Ethanol
Aldehyde	–CHO	-al	Ethanal
Ketone	>C=O	-one	Propanone
Carboxylic acid	–COOH	-oic acid	Ethanoic acid
Ether	R–O–R	alkoxyalkane	Methoxymethane
Ester	–COO–	-oate	Ethyl ethanoate

- **IUPAC rules:** (1) choose the **longest chain** containing the functional group; (2) number so the functional group gets the **smallest number**; (3) name the branches as prefixes.

Isomerism

- **Isomers:** same molecular formula, different structures.
- **Chain isomers:** C_4H_{10} → butane and 2-methylpropane.
- **Position isomers:** C_4H_8 → but-1-ene and but-2-ene.
- **Functional isomers (extra):** ethanol CH_3CH_2OH and methoxymethane CH_3OCH_3, both C_2H_6O.

Formula	Isomers
C_4H_{10}	2
C_5H_{12}	3
C_4H_8 (chain / position)	but-1-ene, but-2-ene, 2-methylpropene

Alkanes

Lab prep (methane)	$CH_3COONa + NaOH \rightarrow CH_4 + Na_2CO_3$ (CaO)
Ethane	$C_2H_5COONa + NaOH \rightarrow C_2H_6 + Na_2CO_3$
From iodomethane	$CH_3I + 2[H] \rightarrow CH_4 + HI$
Complete burning	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
Incomplete	$2CH_4 + 3O_2 \rightarrow 2CO + 4H_2O$
Substitution	$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ (sunlight)

- **Substitution** is the typical reaction of alkanes. Repeats to give CH_2Cl_2, $CHCl_3$, CCl_4.

Alkenes & alkynes

Ethene (lab)	$C_2H_5OH \rightarrow C_2H_4 + H_2O$ (conc. H_2SO_4, 170 °C)
Ethyne	$CaC_2 + 2H_2O \rightarrow C_2H_2 + Ca(OH)_2$

- **Addition reactions** (unsaturated):

Hydrogenation	$C_2H_4 + H_2 \rightarrow C_2H_6$ (Ni, 200 °C)
Bromine water	$C_2H_4 + Br_2 \rightarrow C_2H_4Br_2$ (decolourises)
HCl	$C_2H_4 + HCl \rightarrow C_2H_5Cl$
Ethyne + $2H_2$	$C_2H_2 + 2H_2 \rightarrow C_2H_6$ (Ni)

- **Test for unsaturation:** decolourises bromine water or alkaline $KMnO_4$.

Alcohols & carboxylic acids

Ethanol: oxidation	$C_2H_5OH \rightarrow CH_3COOH$ ($K_2Cr_2O_7$ / H^+)
With Na	$2C_2H_5OH + 2Na \rightarrow 2C_2H_5ONa + H_2$
Esterification	$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$ (conc. H_2SO_4)
Ethanoic + NaOH	$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$
Ethanoic + Na_2CO_3	$2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2$

- **Ester:** fruity smell. **Glacial acetic acid:** pure ethanoic acid (m.p. 16.6 °C).

11 Gas tests, colours & score-savers

Gas identification

Gas	Test	Observation
H_2	Burning splint	Pops
O_2	Glowing splint	Relights
CO_2	Limewater	Turns milky
SO_2	Acidified $K_2Cr_2O_7$	Orange → green
NH_3	Moist red litmus / HCl rod	Blue / white fumes
HCl	NH_4OH rod	White fumes
Cl_2	Moist blue litmus	Turns red, then bleaches
H_2S	Lead acetate paper	Turns black
NO_2	Colour / smell	Brown gas
H_2O	Anhydrous $CuSO_4$	White → blue

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